

Problem G. Crayon

Time Limit	1000 ms
Mem Limit	1572864 kB
Code Length Limit	50000 B
OS	Linux

Background



Mary love painting so much, but as we know she can't draw very well. There is no one appreciate her works, so she come up with a puzzle with herself...

Description

There are only one case in each input file, the first line is a integer N ($N \leq 1,000,00$) denoted the total operations executed by Mary. Then following N lines, each line is one of the folling operations.

- **D L R:** draw a segment $[L, R]$, $1 \leq L \leq R \leq 1,000,000,000$
- **C I:** clear the i th added segment. It's guaranteed that the every added segment will be cleared only once.
- **Q L R:** query the number of segment sharing at least a common point with interval $[L, R]$. $1 \leq L \leq R \leq 1,000,000,000$.

Input

n

(then following n operations ...)

Output

(for each query, print the result on a single line ...)

Example

Input	Output
6 D 1 3 D 2 4 Q 2 3 D 2 4 C 2 Q 2 3	2 2